

Two smooth counterexamples for continuous semigroups and semicocycles

Negative answers to two questions in arXiv:1807.09965

Autonomous proof search, lane 18

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Status. Candidate full counterexamples, likely valid; expert review is requested. The examples answer the two general continuous/smooth questions negatively. They do not concern the narrower positive holomorphic-hyperbolic setting.

1 Source questions

Elin, Jacobzon, and Katriel study continuous and holomorphic semicocycles on domains in real or complex Banach spaces [1]. A bounded set $S \subset \mathcal{D}$ lies *strictly inside* $\mathcal{D}$ when $\inf_{x \in S} \text{dist}(x, \partial\mathcal{D}) > 0$. Their Definition 2.4 says that a semigroup $\mathcal{F} = \{F_t\}_{t \geq 0}$ acts strictly inside $\mathcal{D}$ when

$$\{F_t(x) : x \in S, 0 \leq t \leq t_0\}$$

lies strictly inside $\mathcal{D}$ for every strictly-inside S and every $t_0 > 0$. Immediately afterward they state that, in infinite dimension, they do not know whether every continuous semigroup acts strictly inside. The statement appears on source PDF page 5:

inside $\mathcal{D}$.

Recall in this connection that one of the surprising features of the infinite-dimensional holomorphy is that the inclusion $f \in \text{Hol}(\mathcal{D}, Y)$ does not imply that f is bounded on all subsets strictly inside $\mathcal{D}$ (see [12, 14, 20]).

The following notion concerns the flow's spread over finite times.

Definition 2.4. Let $\mathcal{F} = \{F_t(x)\}_{t \geq 0}$ be a semigroup on a domain $\mathcal{D}$. We say that $\mathcal{F}$ acts strictly inside $\mathcal{D}$ if for every subset $\mathcal{D}^*$ strictly inside $\mathcal{D}$ and every $t_0 > 0$ the set $\{F_t(x) : x \in \mathcal{D}^*, t \in [0, t_0]\}$ lies strictly inside $\mathcal{D}$.

In the case where X is finite-dimensional, any semigroup acts strictly inside by compactness. On the other hand, we do not know whether in the infinite-dimensional case every semigroup $\mathcal{F} \in C(\mathcal{D})$ acts strictly inside. Clearly, if $\mathcal{D}$ is the union of $\mathcal{F}$ -invariant subsets each one of which is strictly inside $\mathcal{D}$, then $\mathcal{F}$ acts strictly inside. Moreover, it can be shown that if a semigroup $\mathcal{F}$ consists of holomorphic self-mappings of a domain $\mathcal{D}$ equipped

For a family $\{G_t : \mathcal{D} \rightarrow Y\}_{t \geq 0}$, T-continuity at t_0 means

$$\sup_{x \in S} \|G_t(x) - G_{t_0}(x)\| \longrightarrow 0 \quad (t \rightarrow t_0)$$

for every S strictly inside $\mathcal{D}$. Before Proposition 3.3, the authors ask whether T -continuity of a semicycle at $t_0 = 0$ implies T -continuity for all $t_0 \geq 0$. Their Proposition 3.3(ii) proves this under two extra assumptions: the base semigroup acts strictly inside and every semicycle map is bounded on sets strictly inside. The open statement appears on source PDF page 10:

with respect to n . Therefore, this semicycle is not T -continuous, while its uniform joint continuity is obvious.

Considering the extended semigroup $\tilde{\mathcal{F}}$ defined in Proposition 3.1 the relationship between uniform joint continuity and T -continuity of a semicycle and those of the extended semigroup defined by it follows easily from the definition:

Proposition 3.2. *Let $\{\Gamma_t\}_{t \geq 0} \subset C(\mathcal{D}, \mathcal{A})$ be a semicycle over some semigroup $\mathcal{F} \subset C(\mathcal{D})$. The semigroup $\tilde{\mathcal{F}}$ is uniformly jointly continuous (respectively, T -continuous) if and only if both $\mathcal{F}$ and $\{\Gamma_t\}_{t \geq 0}$ are.*

One might conjecture that uniform joint continuity (or T -continuity) of a semicycle at $t_0 = 0$ will imply the same property for all $t_0 \geq 0$. Unfortunately, we do not know whether such a conclusion concerning the T -continuity is true in general. We can show it only under some restrictions which are always satisfied when X is finite-dimensional.

Proposition 3.3. *Let $\{\Gamma_t\}_{t \geq 0} \subset C(\mathcal{D}, \mathcal{A})$ be a semicycle over a semigroup $\mathcal{F} \subset C(\mathcal{D})$.*

- (i) *If the semicycle is UJC at $t = 0$ then it is UJC.*
- (ii) *Assume in addition that $\{F_t\}_{t > 0}$ acts strictly inside $\mathcal{D}$ and every map-*

2 Proof intuition

Infinite dimension allows a smooth function to be unbounded even on a set that stays a fixed positive distance from the boundary. We build such a function g explicitly on ℓ_2 , with $g(\frac{1}{2}e_n) = n$. For each fixed u , the vertical slice of the Hilbert ball is an interval. The change of coordinate $\sigma = R(u) \tanh y$ identifies that interval with the whole real line. We let the semigroup translate y at speed $g(u)$. The basis points are therefore sent arbitrarily close to the boundary by time one, although they began deep inside the ball.

For the second question, a bounded coboundary $\Gamma_t = \exp(q \circ F_t - q)$ automatically obeys the semicycle identity. Choose q to be a smooth step whose threshold is tuned to the time-one translation. On any fixed strictly-inside set, small-time motion is uniformly invisible: either the speed is bounded, or both sides of the step remain in its flat zero region. At time one, however, all basis points sit exactly at the threshold, and an arbitrarily small further time activates a sufficiently fast coordinate. This destroys T -continuity at time one while keeping every Γ_t globally bounded.

3 An analytic speed unbounded on an interior sequence

Let $H = \ell_2(\mathbb{R})$ with standard basis $(e_k)_{k \geq 1}$ and define

$$g(u) = \sum_{k=1}^{\infty} k(2u_k)^{2k}, \quad u = (u_k) \in H. \quad (1)$$

Lemma 1. *The series in (1) defines a nonnegative real-analytic function on H . Moreover,*

$$g\left(\frac{1}{2}e_n\right) = n \quad (n \geq 1).$$

Proof. Fix $u \in H$. Since $u_k \rightarrow 0$, choose N and $q_0 < 1$ so that $2|u_k| \leq q_0$ for $k > N$. Choose $\delta > 0$ with $q := q_0 + 2\delta < 1$. If $\|v - u\|_2 < \delta$, then for $k > N$,

$$2|v_k| \leq 2|u_k| + 2\|v - u\|_2 < q.$$

Hence the tail of (1) is uniformly dominated on that neighborhood by $\sum_{k>N} kq^{2k} < \infty$. The identical estimate holds for complex $v \in \ell_2(\mathbb{C})$ near the real point u . Thus the coordinate polynomials $k(2v_k)^{2k}$ converge locally uniformly to a holomorphic function on the complexification. Its restriction to H is real analytic. Nonnegativity is clear, and at $u = \frac{1}{2}e_n$ only the n th summand is nonzero, with value n . $\square$

4 The two counterexamples

Let

$$X = H \oplus_2 \mathbb{R}, \quad \mathcal{D} = B_X = \{(u, \sigma) : \|u\|_2^2 + \sigma^2 < 1\}.$$

For $\|u\|_2 < 1$, put

$$R(u) = \sqrt{1 - \|u\|_2^2}, \quad y(u, \sigma) = \operatorname{artanh}\left(\frac{\sigma}{R(u)}\right).$$

Thus $|\sigma| < R(u)$, and $\sigma \mapsto y(u, \sigma)$ is a real-analytic diffeomorphism from the vertical slice $(-R(u), R(u))$ onto $\mathbb{R}$.

For $t \geq 0$, define

$$F_t(u, \sigma) = (u, R(u) \tanh(y(u, \sigma) + tg(u))). \quad (2)$$

Choose a C^∞ nondecreasing function $Q : \mathbb{R} \rightarrow [0, 1]$ satisfying

$$Q(r) = 0 \ (r \leq 0), \quad Q(r) = 1 \ (r \geq 1). \quad (3)$$

For example, one may use the standard transition constructed from $\theta(r) = 0$ for $r \leq 0$ and $\theta(r) = e^{-1/r}$ for $r > 0$: $Q(r) = \theta(r)/(\theta(r) + \theta(1 - r))$. Set

$$q(u, \sigma) = Q(y(u, \sigma) - g(u)), \quad \Gamma_t(x) = \exp(q(F_t(x)) - q(x)). \quad (4)$$

Theorem 1. *The following assertions hold.*

1. *The family $\mathcal{F} = \{F_t\}_{t \geq 0}$ is a semigroup of real-analytic self-maps of the open unit ball of an infinite-dimensional real Hilbert space, but $\mathcal{F}$ does not act strictly inside $\mathcal{D}$.*
2. *With the unital complex Banach algebra $\mathcal{A} = \mathbb{C}$, the family $\{\Gamma_t\}_{t \geq 0}$ is a globally bounded C^∞ scalar semicycle over $\mathcal{F}$. It is T -continuous at $t = 0$ but is not T -continuous at $t = 1$. In fact $e^{-1} \leq \Gamma_t(x) \leq e$ for every $t \geq 0$ and $x \in \mathcal{D}$.*

Consequently, both general questions above have negative answers.

Proof. Because $|\tanh r| < 1$ for real r , formula (2) maps $\mathcal{D}$ into $\mathcal{D}$. It is real analytic by Lemma 1. In the y coordinate it is simply

$$(u, y) \mapsto (u, y + tg(u)).$$

Therefore F_0 is the identity and $F_t \circ F_s = F_{t+s}$. Also $F_t(x) \rightarrow x$ as $t \rightarrow 0^+$ for every x , so $\mathcal{F}$ is a continuous semigroup in the precise sense used in [1].

Now put

$$p_n = \left(\frac{1}{2}e_n, 0\right) \in \mathcal{D}.$$

Every p_n has norm $1/2$. Thus $P = \{p_n : n \geq 1\}$ is bounded and $\text{dist}(P, \partial\mathcal{D}) = 1/2$, so P lies strictly inside $\mathcal{D}$. At p_n we have $R(\frac{1}{2}e_n) = \sqrt{3}/2$, $y(p_n) = 0$, and $g(\frac{1}{2}e_n) = n$. Consequently,

$$F_1(p_n) = \left(\frac{1}{2}e_n, \frac{\sqrt{3}}{2} \tanh n\right), \quad \|F_1(p_n)\|^2 = \frac{1}{4} + \frac{3}{4} \tanh^2 n \rightarrow 1.$$

Thus $\inf_n \text{dist}(F_1(p_n), \partial\mathcal{D}) = 0$. The finite-time orbit set for P and $t_0 = 1$ is not strictly inside $\mathcal{D}$, proving part 1.

The function q is smooth and $0 \leq q \leq 1$. The coboundary formula gives $\Gamma_0 = 1$, and for $s, t \geq 0$,

$$\begin{aligned} \Gamma_t(F_s x) \Gamma_s(x) &= \exp(q(F_t F_s x) - q(F_s x) + q(F_s x) - q(x)) \\ &= \Gamma_{t+s}(x). \end{aligned}$$

Pointwise time continuity follows from that of F_t , so this is a smooth scalar semicocycle in the source sense. Since $q(F_t x) - q(x) \in [-1, 1]$, $e^{-1} \leq \Gamma_t(x) \leq e$ globally.

It remains to prove the two time-continuity claims. Let S be any subset strictly inside the unit ball. There is $r < 1$ such that $\|x\| \leq r$ for all $x = (u, \sigma) \in S$. Writing $a = \|u\|_2^2$, we have

$$\left| \frac{\sigma}{R(u)} \right|^2 \leq \frac{r^2 - a}{1 - a} \leq r^2.$$

Hence

$$|y(x)| \leq M := \text{artanh } r \quad (x \in S). \quad (5)$$

Let $L = \|Q'\|_\infty$. Since F_h adds $hg(u)$ to y ,

$$\log \Gamma_h(u, \sigma) = Q(y - (1 - h)g(u)) - Q(y - g(u)). \quad (6)$$

For $0 < h \leq 1/2$, split into two cases. If $g(u) \geq 2M$, both arguments in (6) are at most zero, so the difference vanishes. If $g(u) \leq 2M$, the mean value theorem gives

$$|\log \Gamma_h(u, \sigma)| \leq Lhg(u) \leq 2LMh.$$

Thus $\log \Gamma_h \rightarrow 0$, and hence $\Gamma_h \rightarrow 1$, uniformly on every strictly-inside S . The semicocycle is T-continuous at zero.

Finally consider the same strictly-inside set $P = \{p_n\}$. At p_n , $y = 0$, $g = n$, and therefore $q(p_n) = Q(-n) = 0$. At time one the new y -value is n , so $q(F_1 p_n) = Q(0) = 0$ and $\Gamma_1(p_n) = 1$. But for every $h > 0$ and every integer $n \geq 1/h$,

$$q(F_{1+h} p_n) = Q(hn) = 1, \quad \Gamma_{1+h}(p_n) = e.$$

It follows that

$$\sup_{x \in P} |\Gamma_{1+h}(x) - \Gamma_1(x)| \geq e - 1 \quad (h > 0).$$

Therefore T-continuity fails at $t = 1$, proving part 2. $\square$

5 Consequences, limitations, and novelty check

Part 1 directly disproves the proposed general infinite-dimensional strict-inside property, even for a bounded convex domain in a real Hilbert space and a real-analytic semigroup. Part 2 shows that the strict-inside assumption in Proposition 3.3(ii) cannot simply be deleted, even when every semicocycle map is globally bounded and all data are smooth.

The construction is deliberately real. It does not contradict the source’s observation that semigroups of holomorphic self-maps on domains equipped with the hyperbolic metric act strictly inside. No assertion is made about a holomorphic analogue of this example on a complex hyperbolic domain.

Before promotion, a bounded literature search was performed on 11 August 2026 over arXiv and general scholarly web indexes. Queries included the exact paper title and arXiv id, the two quoted open-question sentences, and combinations of “semigroup acts strictly inside”, “infinite-dimensional Banach”, and “T-continuity at time zero”. The search found the source paper and neighboring literature but no later paper explicitly answering either question and no matching counterexample. This is a bounded novelty check, not a comprehensive priority claim.

Human-review recommendation. Review is recommended at high priority. The principal points to check are the local uniform convergence on the complexification in Lemma 1, the global definition of the Hilbert-ball coordinate flow, and the two-case estimate following (6).

References

- [1] M. Elin, F. Jacobzon, and G. Katriel, *Continuous and holomorphic semicocycles in Banach spaces*, J. Evol. Equ. **19** (2019), 1199–1221; arXiv:1807.09965; DOI: 10.1007/s00028-019-00509-5.